Google Cloud H2S HACKATHON

AI Builder Cup

Build in JAPAC.
Build for the world.



Team Details

a. Team name: **Techno Crackers**

b. Team leader name: **Venket Raj S** (Team: Venket Raj S · Malik Chandra Pandian · Arun Veerappan · Nagaraj — 4 working professionals in JAPAC, no students)

c. Problem Statement: **BFSI: Intelligent Risk, Fraud & Financial Experiences**

CatSight — Event-aware contract intelligence for catastrophe reinsurance: After a typhoon or earthquake, a reinsurer cannot say quickly or defensibly how much it owes because the answer depends on contract clauses buried in PDF treaty wordings.

Brief about the idea

CatSight tells a reinsurer, **in under a minute** after a typhoon or earthquake, which contracts pay, how much, and which clauses could change that, with every quote checked against the contract page. The money is on screen in about a second.

The problem, in people terms

- After Typhoon Jebi, insurers paid **857,284 claims (¥1,067.8bn)** (GIAJ) to families and businesses.
- They recover much of it from **reinsurers**, under 50-page legal contracts called **treaties**.
- Working that out takes an estimated **30–60 analyst-hours**, and unclear clauses (wind vs. storm-surge flooding) lead to disputes. Slow payouts mean slower rebuilding.

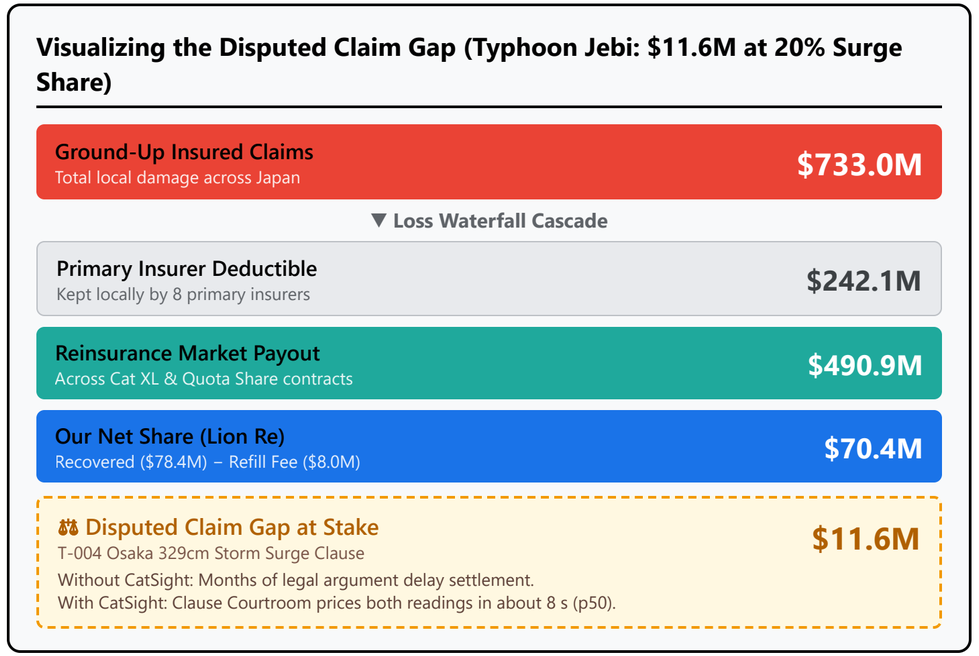
Real disasters, fictional contracts. Real treaties are confidential, so we replay real public hazard data over **24 synthetic treaties for 8 fictional insurers** in Japan, the Philippines and Taiwan.

Source	Time to arrive	What it gives you
Catastrophe-model estimate	**4–20 days**	A market-wide range, not your contracts
PERILS (Asia loss index)	**6 weeks**	An industry total, not treaty payouts
Verisk PCS Japan	**up to 90 days**	Market index, events ≥ $2bn only
Manual spreadsheet review	**2–4 days**	Slow; easy to miss endorsements
CatSight	**0.7 s money · 8–43 s full check**	**Your treaties, priced, with checked quotes**

Hazard data: NOAA IBTrACS (typhoon tracks), USGS ShakeMap (earthquakes).

Opportunities

- **How is it different?**
  - Catastrophe models estimate the loss but stop before your contracts. Document AI reads contracts but not the disaster. **CatSight joins the two, in seconds.**
  - Works **alongside** licensed models (Moody's RMS, Verisk): their footprint can plug in.
- **How does it solve the problem?**
  - **Tested code does every calculation:** money on screen in **0.7 s** (median). The AI never does maths.
  - **Gemini finds the clauses that matter;** every quote is matched word-for-word, then checked by a second model.
- **What's unique, and the social impact**
  - **Clause Courtroom:** two AI lawyers argue an unclear clause; the engine prices both readings (**$11.6M** at stake on the recorded run).
  - **Reads scanned, stamped endorsements** that have no text layer.
  - **91%** of APAC disaster losses ($65B) are uninsured; **48%** of insurers lack catastrophe models. Faster, clearer settlements help communities rebuild.



List of features offered by the solution

Feature	What it gives the analyst	How it stays trustworthy
1. Who pays what	Matches all 24 contracts by peril, place and dates (with a reason for each of the 11 that don't respond); payout per layer, coverage used, refill fees, and where the insurer, all reinsurers and Lion Re end up.	Tested code; exact match with a hand-worked example.
2. "How was this calculated?"	Each contract step by step: wind or shaking, damage %, insured value × damage % = loss, and each layer's formula with its citation.	Every number can be inspected.
3. Clause flags & scanned pages	Flags clauses that bite on *this* event (T-004 excludes storm surge; Jebi's 329 cm Osaka tide) and reads a scanned endorsement (168 h replaces 72 h).	Two-step quote check; failures shown as withheld; scanned pages badged *transcribed*.
4. Clause Courtroom	Two AI lawyers argue opposite readings in parallel, from checked quotes; the engine prices both; a neutral arbiter summarises.	Issue-spotting only; the analyst records the decision.
5. What-if & follow-up questions	Try a labelled hypothetical storm window; ask plain questions about the contracts and results.	Bounded inputs; questions screened by a local filter + **Google Cloud Model Armor**.
6. Replay, tour & reports	6-second free replay of a real run, 9-stop guided tour, colour-blind-safe themes, Word & Markdown reports.	Every report carries the method version, check counts and disclaimers.

Process flow diagram or Use-case diagram

Workflow 1: live analysis (ADK 2.11 `Workflow`)

1. event_intel
Loads the real storm track or earthquake shaking map, and the event's timeline

▼

2. exposure_analyst (no AI: tested Python)
Matches 24 treaties → loss by region → payout per layer. Money on screen in ~0.7 s

▼

3. wording_checker (Gemini 3.5 Flash-Lite)
Searches each treaty's clauses (Firestore vector search) → candidate flags with page, clause and quote

▼

4. verifier ⇄ reviser (one retry)
Word-for-word quote match, then Gemini 3.8 Flash checks the quote supports the claim → one retry → else withheld

▼

5. reporter (Gemini 3.5 Flash-Lite + number check)
Writes the memo from verified facts; code checks every number against the engine

Workflow 2: Clause Courtroom (parallel)

1. brief
Picks the most material verified exclusion flag and the clauses around it

▼

2a. cedent_counsel
Flash-Lite · argues the insurer's reading + excluded share (0–50%)

2b. reinsurer_counsel
Flash-Lite · argues the reinsurers' reading + excluded share (0–50%)

▼ Join + quote & support check (one retry per side)

3. price (tested Python)
Prices both readings through the treaty layers; money at stake for each party

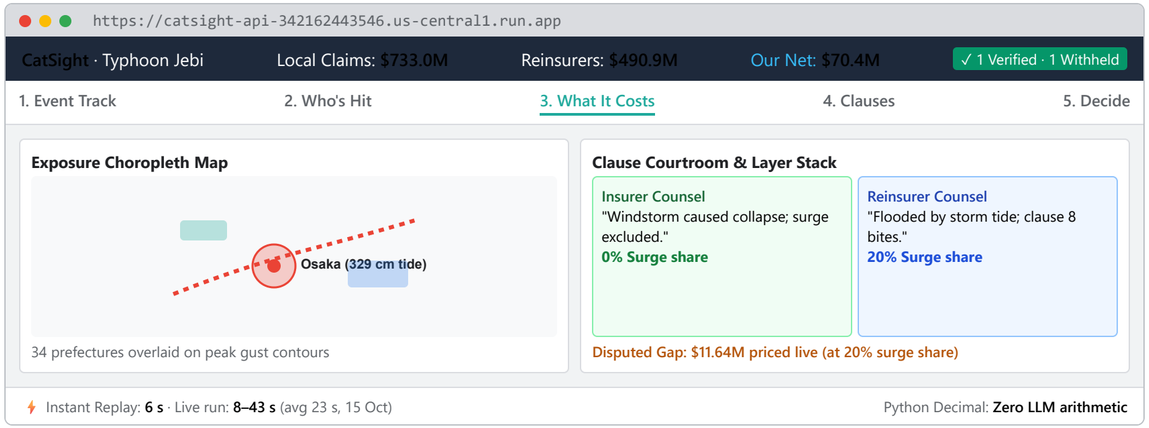
▼

4. arbiter (Gemini 3.8 Flash) → the analyst decides
Summarises both sides without deciding; the analyst records a position

Wireframes/Mock diagrams of the proposed solution (optional)

Analyst Cockpit: 5-Step Progressive UI Wireframe

React 18 + SSE Stream



Trust Architecture: 2-Stage Verification Funnel

Every quote checked

1. Candidate AI Clause Flag / Counsel Argument
Model proposes: `treaty_id`, `page`, `clause_no`, and verbatim `quote`

▼ Filter 1

Stage 1: Deterministic Word-Aligned Quote Matcher — 0 ms Python
Normalised substring + ellipsis (`...`) matcher against Firestore treaty PDF text chunk. Catches 40/40 corrupted quotes; tags scanned endorsements as `transcribed`.

▼ Filter 2

Stage 2: Semantic Support Check (Gemini 3.8 Flash) — Model reasoning
Verifies quote + event timeline (`start`, `end`, `duration_h`) + JMA facts directly support the exclusion. 1 revision retry loop on failure.

✓ Verified Citation
Rendered in UI with clickable source PDF page pill & pricing impact

⊘ Visibly Withheld
Suppressed from findings; shown as `n verified · k withheld`

Architecture diagram of the proposed solution

Analyst / judge browser (React 18 + TypeScript)
One origin over HTTPS · results stream live (server-sent events)

▼

Google Cloud Run — `catsight-api` (us-central1)
One container: UI + FastAPI · ADK 2.11 `Workflow` graphs · tested Python loss engine · least-privilege service account · max 3 instances · hourly & daily caps

▼

Gemini Enterprise Agent Platform
(formerly Vertex AI)
- 3.5 Flash-Lite
- 3.8 Flash
- embedding-001

Firestore
- Treaty book & events
- 252 clause vectors
- Saved runs

Cloud Storage
- Treaty PDFs
- Stamped endorsements

Model Armor
- Screens follow-up questions & answers

▼

Cloud Logging · Trace · Monitoring · Log Analytics
A log line per stage · guardrail metrics (citations withheld, lawyers dropped, unsupported figures) · live dashboard

Key decisions

- **Money first, AI second:** tested code prices the loss before any AI call (0.7 s). The AI never calculates.
- **Rules are data, not code:** treaty terms, damage curves and market rules live in versioned data files; the version in force is stamped on every run.
- **One service, no keys:** UI and API on one Cloud Run service; a dedicated service account, no API keys.
- **Ready for judging:** pre-computed searches (no embedding calls on a cold start), one warm instance, and rate limits that fall back to the recorded replay.

Technologies to be used in the solution

Technology / Service	Role	Why we chose it
Gemini 3.5 Flash-Lite Gemini Enterprise Agent Platform	Agents & page reading	Fast, low-cost structured output for wording checks, the two Courtroom lawyers, report text and the scanned page.
Gemini 3.8 Flash Gemini Enterprise Agent Platform	Checking & arbiter	Stronger reasoning to check that each quote supports its claim, and to summarise the Courtroom fairly.
Google ADK 2.11	Agent workflows	Graph `Workflow`: a 5-step analysis with a bounded retry, and a Courtroom with two lawyers in parallel (`JoinNode`).
Firestore + Vector Search `gemini-embedding-001`	Data & clause search	Treaty book, 252 clause vectors filtered by treaty (186/186 governing clauses found), runs and audit trail.
Google Cloud Model Armor	Question screening	Screens follow-up questions and answers for prompt injection, jailbreaks and harmful content, after a local filter.
Cloud Run & Cloud Storage	Hosting & PDFs	One container for UI + API; scale-to-zero by default, one warm instance during judging; Storage holds the treaty PDFs.
Cloud Logging, Trace & Monitoring	Observability	A log line per stage, per-agent traces, and a live dashboard built from log-based metrics.
Python 3.12 · FastAPI · React 18	Engine & UI	Tested loss engine (870+ offline + 13 live tests), Word report generator, accessible TypeScript UI.

Estimated implementation cost (optional)

$0.010
Median AI cost per live analysis ($0.008–$0.014)

$0.005
Median AI cost per Clause Courtroom run ($0.005–$0.006)

$0.000
Instant Replay (a recorded real run, no AI calls)

$0.003
Follow-up question (from token counts)

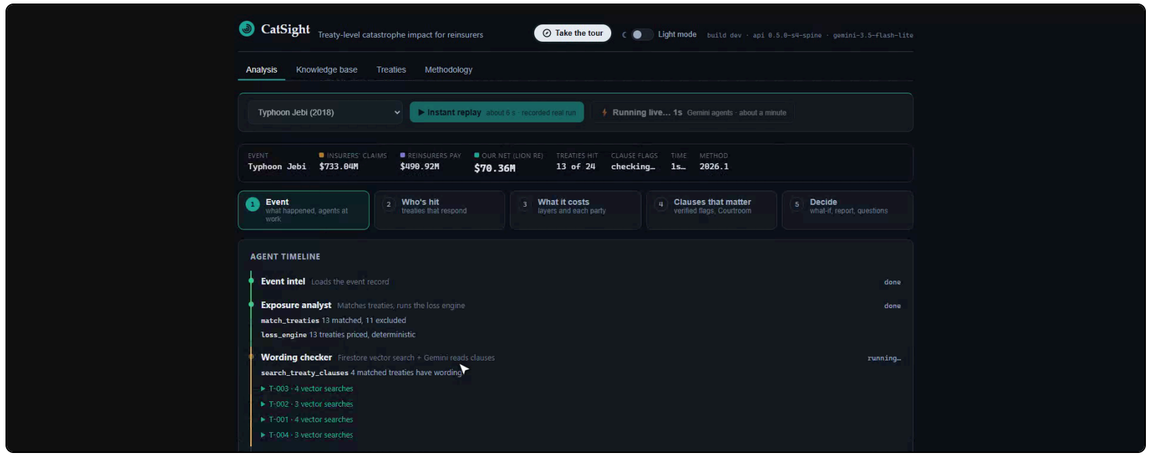
Measured Cloud & Token Breakdown

Workload	Model / Service	Measured Cost
One-time ingestion (8 PDF wordings, 252 chunks)	`gemini-embedding-001` + `3.5-flash-lite`	< **$0.08 total** (57 s runtime)
Live event analysis (13 responding treaties)	`3.5-flash-lite` + `3.8-flash` verifier	**$0.0097 p50** ($0.0081–$0.0140)
Clause Courtroom (parallel counsel + arbiter)	2× `3.5-flash-lite` + `3.8-flash`	**$0.0052 p50** ($0.0047–$0.0064)
Cloud Run + Firestore	`min-instances 0` by default (1 during judging), `max 3`	**~$0 idle** at zero instances · pennies per active hour

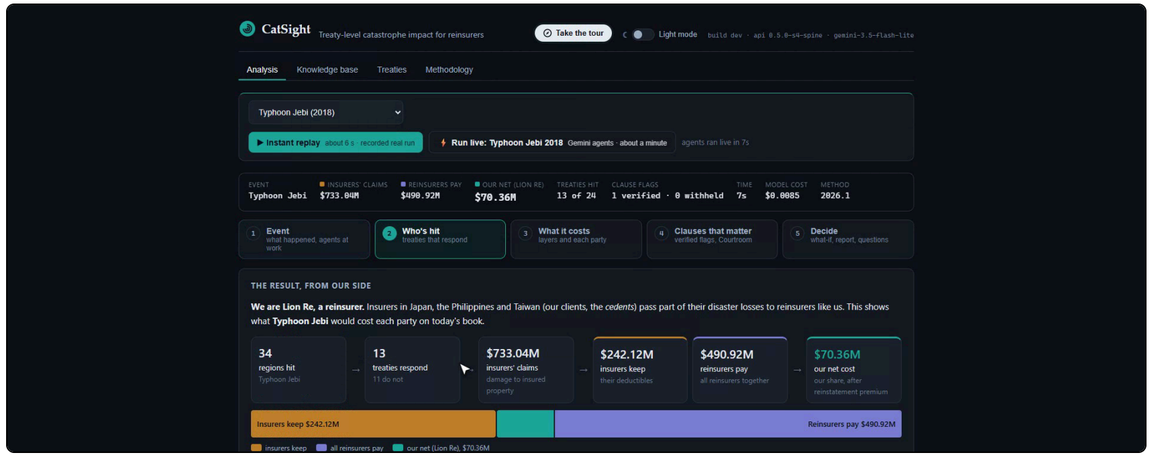
Annual Run-Rate vs. Manual Analyst Effort

- **A mid-size reinsurer:** 8 major events + 20 smaller checks a year (≈ 500 runs) → **~$7/year of Gemini cost**, plus ~$25–$40/month for Cloud Run and Firestore when active.
- **Analyst time saved (estimate):** 30–60 hours per event down to 3.5–8 hours → **165–675 hours a year** (about $12k–$100k; labelled *simulated*).
- **Cost guards:** 20 live runs/hour per visitor, 300/day cap, automatic fallback to Instant Replay, billing budget alerts.

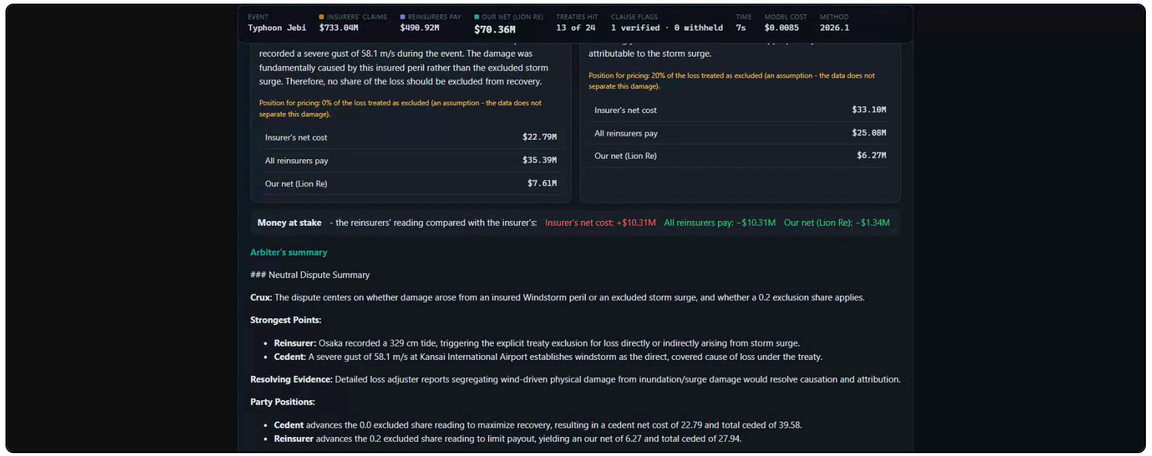
Snapshots of the prototype



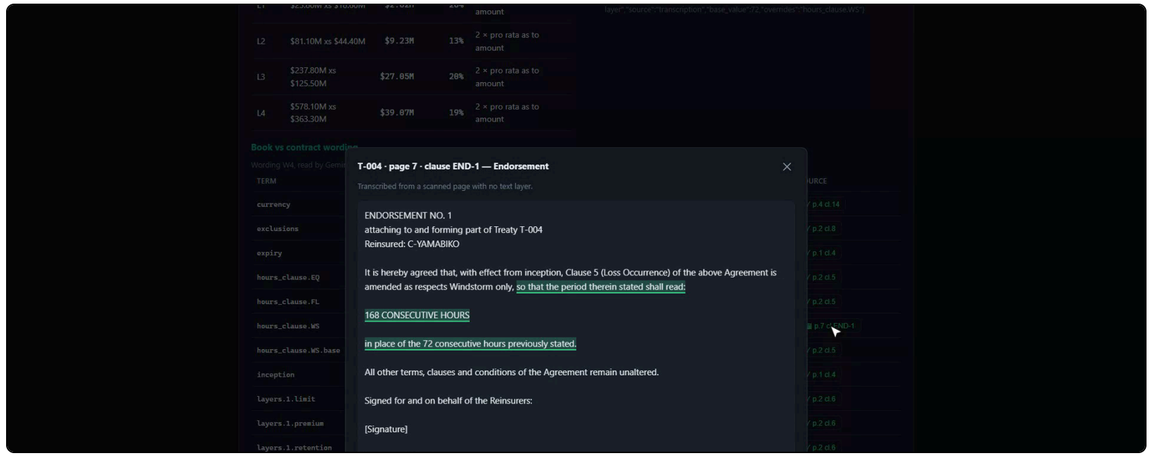
1. Live Agent Streaming: ADK 2.11 stages stream over SSE; deterministic loss figures arrive in ~0.7 s while Gemini retrieves & verifies treaty clauses.



2. Three-Party Loss Waterfall: 13 of 24 treaties respond on Typhoon Jebi ($733.04M claims → $242.12M insurer retention, $490.92M reinsurers, $70.36M Lion Re net).



3. Clause Courtroom (Jebi T-004 Storm-Surge Exclusion): Parallel counsel argue 0% vs 20% surge share; engine prices both (Ceded −$11.64M, Insurer +$10.31M, Reinsurers −$10.31M, Lion Re −$1.34M).



4. Scanned Endorsement Detective (T-004 p.7): Gemini transcribes a rasterised stamped endorsement with no text layer (168 hours replaces 72 hours) and badges it `transcribed`.

Prototype Performance report/Benchmarking

0.7 s
Money on screen (median; 0.5–1.1 s over 10 live runs)

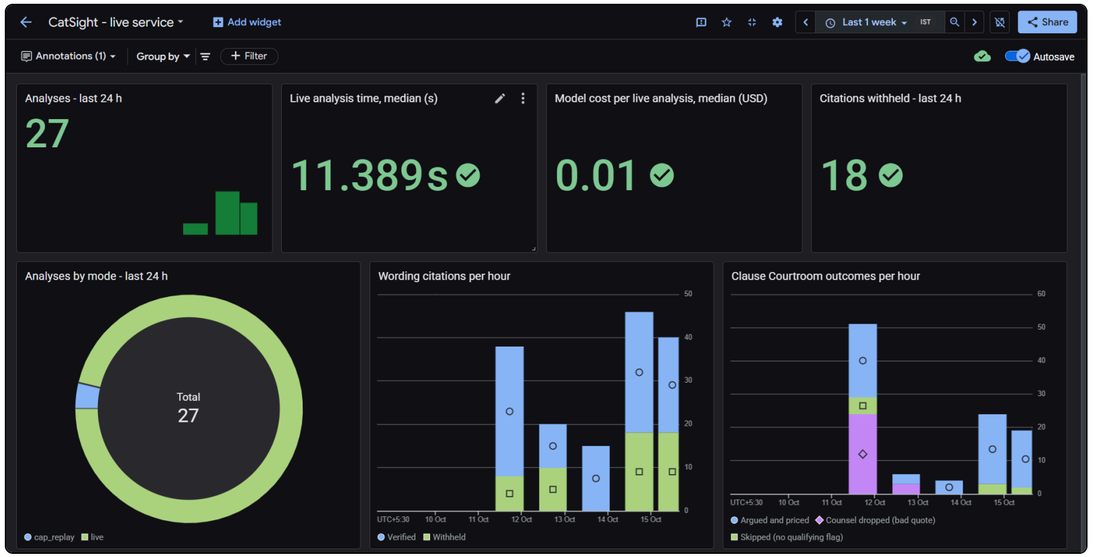
23 s
Full analysis with checked quotes (average of 3 live runs, 15 Oct; 8–43 s). Benchmark median 11.3 s

42 s
Clause Courtroom, a separate click (average of the same 3 runs; 10–70 s)

99.5%
Treaty terms extracted (185/186 fields, in 57 s vs ~25 min)

870+
Automated tests (870+ offline + 13 live; CI on GitHub)

What we measured (`docs/RESULTS.md`)	Result
Loss engine vs. hand-worked example every layer, reinstatement and quota-share formula	**100% exact**
Clause search & citation audit 1 scanned page transcribed; 1 unverified withheld	**186/186 found · 185/186 verified**
Clause Courtroom, live skipped in 2 when the T-004 flag was withheld; $10.3M–$26.0M at stake priced live	**Ran in 8 of 10; both sides verified in all 8**
Bad-quote rejection (offline test suite)	**40/40 rejected, 0 false withholds**
Untraceable numbers in AI-written text	**0 across all live runs**



The live-service dashboard in Cloud Monitoring (screenshot, 15 Oct 2026). "Citations withheld" counts quotes the checker refused to show: the guardrail working, not an error.

Honest limits: extraction prompts were tuned on the same 8 synthetic wordings (99.5% is an upper bound; a blind set is next). Speed varies: slow runs spend their time re-checking a quote that failed its first check, and one Courtroom on 15 Oct took 278 s. Analyst-time savings are labelled *simulated estimates*.

Additional Details/Future Development (if any)

Why pricing each party matters

1. Jebi Courtroom (T-004 storm-surge exclusion, p.2 cl.8):
At a 20% surge share (the recorded run), reinsurance payouts fall **−$11.64M** ($39.58M → $27.94M); net of refill fees, reinsurers save **$10.31M** and the insurer carries **+$10.31M**; Lion Re saves **$1.34M**. The demo video shows a separate 50% run: **−$29.09M** ceded, Lion Re **−$5.16M**.

2. Hours clause (worked example, What-if):
Splitting one $54.4M event into two ($35.0M + $19.4M) cuts reinsurers' recoveries by **−22.5% (−$10.0M)**, yet **raises Lion Re's loss +7.4%** ($6.82M → $7.33M), because the split moves loss into the layer where Lion Re's share is largest.

Team Techno Crackers — Roles

• **Venket Raj S (Lead):** Architecture, Cloud Run / Firestore / IAM deployment & observability
• **Malik Chandra Pandian:** Reinsurance domain modelling, `Decimal` loss engine & worked examples
• **Arun Veerappan:** ADK 2.11 workflows, RAG extraction, citation verifier & Clause Courtroom
• **Nagaraj:** React 18 + TypeScript + custom SVG choropleth UI, 5-step analyst workflow, tour & exports

Future Development Roadmap (Designed / Investigated)

- **A real hours-clause dispute:** the 2024 Hualien earthquake (Taiwan). Its real aftershocks ~20 days later are a second occurrence worth ~$11–15M per treaty.
- **Damage curves per country:** calibrated for the Philippines, bringing Typhoon Haiyan back into the demo.
- **Licensed hazard footprints:** Moody's RMS or Verisk behind our hazard interface.
- **Your own contracts:** authenticated treaty upload, and BigQuery for portfolio-scale exposure.
- **Automatic start:** USGS / GDACS disaster alerts via Pub/Sub, plus an A2A/MCP endpoint for other agents.

Provide links to your:

- **GitHub Public Repository**
  https://github.com/venketraj/catsight-ai-buildercup-2026
  Apache-2.0 licence · 870+ offline tests + 13 opt-in live tests · GitHub Actions CI · Full architecture & assumptions in `docs/`.

- **Demo Video Link (under 3 minutes: 2 min 48 s)**
  https://drive.google.com/file/d/14Yv-A-KtVksOgRxOxafZcEKFBUvFumQi/view?usp=sharing
  One live Typhoon Jebi run through all five steps: the Clause Courtroom, a scanned endorsement read by Gemini, What-if, the Word report and a follow-up question.

- **Final Product Link**
  https://catsight-api-342162443546.us-central1.run.app
  Live on Google Cloud Run · No login · **Instant replay** (~6 s) or **Run live** (~23 s on average; the Courtroom ~42 s more) · 9-stop tour · 3 example questions built in · the README has a 2-minute judge guide.

- **Submission Folder (Google Drive)**
  https://drive.google.com/drive/folders/1X5cNQh4aMP-B1JIvIXyyxCL-4anxJhl4?usp=sharing
  The demo video and this deck (PDF).

Disclaimers & Data Attribution: Indicative first view on synthetic data. Not a catastrophe-model output. Not underwriting or claims advice. Issue-spotting only — not legal advice. Jebi, Hagibis, Haiyan and Noto were real events in which people died. Portfolio, insurers and treaty wordings are 100% synthetic; hazard data is public (NOAA IBTrACS v04r01, USGS ShakeMap, Natural Earth).

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Thank you

